



ESP SAFETY PVT. LTD.

Transmittal Cover Page

Sr. No.	Details	Description
	General:	
1	Client:	DE's TECHNICO LIMITED
2	End User:	BERGER GOA
3	Project Name:	BERGER GOA
4	Order Number:	DTL/ ESP SAFETY/FLAME & GAS DETECTOR/SM-561/24-25
5	Technical Lead:	Projects E1
6	Document Name	Technical Documents of Flame Detectors
7	Document Number	Documents
8	Client Reference Number	NA
0	09-01-2025	PE PM DIR
Revision	Dated	Prepared by Reviewed by Approved by

Document Name

Technical Documents of Flame Detectors

Client (Logo)	End User (Logo)

Datasheet of Flame Detector

Technical Documents of Flame Detectors			
Document Number	Documents	Client Reference Number	NA
Sr. No.	Description	Technical Specification	Option/Data (If Applicable)
	General:		
1	Make	ESP	
2	Model	IPES IR3	
3	Quantity	61	
4	Type	Multichannel Infrared Flame Detector	
	Sensor Specificaiton:		
5	Detection Technology	Triple Infrared	
6	Input Voltage	24V DC (18-32 VDC)	
7	Outputs	4-20mA	Yes
		RS-485	Yes
		Relay	Yes
		HART	-
8	Detection Range	Upto 64 meters	
9	Viewing Angle	120° Conical	
10	Response Time	Flame:	<3.0 sec
11	Operating Temperature	-40°C to +85° C	
12	Humidity Range	0-95% RH	
13	Ingress Protection	IP 66	
14	Status Indication	3 colour LED (normal, fault, alarm indication)	
15	Power Consumption	<2.0W - Standby	
		<3.6W - Test Mode	
		<7.0W - Heated Optic	
	Certficaiton & Other:		
16	SIL Rating	SIL-3	
17	Approvals	ATEX/SIL/PESO/FM	
18	Enclosure Material	SS316	Yes
19		Aluminnium	-
20	Ex Marking	Ex d IIC T4 Gb	
21	Cable Entry	M20 (2 Nos.)	
22	Weight	Approx. 6 Kg	
23	Accessories	Cable Gland	Yes
		Blind Plug	Yes
		Sensor Cap	Yes
		Nuts & Bolts (Mounting Accessories)	Yes
		2" Pipe Mounting Kit	-

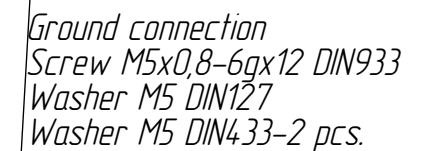


		Canopy	-
	Notes & Comments		
Note 1			
Note 2			
0	09-01-2025	PE	DIR
Revision	Dated	Prepared by	Approved by



GAD of Flame Detector

Ground connection
Stud M5x0,7-6gx20
Nut M5 DIN934-2 pcs.
Washer M5 DIN127
Washer M5 DIN433-2 pcs.



1. Reference dimensions.
2. Not any mechanical damages such as knicks, dents, scratches and blisters are allowed on surfaces labelled with a word "Explosion".
3. Apply a light layer of corrosion-resistant anti-seize lubricant such as CIATIM-221 or equiv to surfaces labelled with a word "Explosion", a sealing ring, contact surfaces of the earthing clip prior assembly.
4. Quantity of complete nonfailed threads of screws not less than 5.
5. Enclosure free volumes are $V_1 = 325 \text{ cm}^3$, $V_2 = 171 \text{ cm}^3$.
6. Test pressure is 0,6 MPa.
7. Epoxy encapsulant Stycast 2651 | Catalyst 9 (Emerson & Cuming)

QAP of Flame Detector

	ESP SAFETY PVT. LTD.	Doc No.
	Quality Assurance Plan (QAP)	ESP/Operation/PFQ1.0
		Rev: 01
		DD: 07.10.2020

OEM	JSC "ELECTRONSTANDART-PRIBOR"
Product Name/ Model Number	Flame Detectors / IPES IR3
Quantity	61 Nos.
Notes & Remarks	

Client Name:	DE's TECHNICO LIMITED
End User:	BERGER GOA
Purchase Order No. :	DTL/ ESP SAFETY/FLAME & GAS DETECTOR/SM-561/24-25
Project Ref/Tender No.:	BERGER GOA
Document No.	QAP

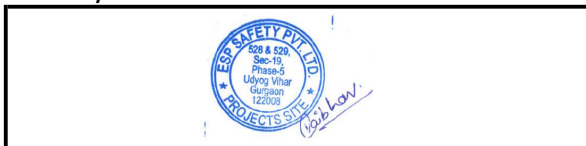
S.No.	Description of inspection	Quantam check	Acceptance Criteria	Documents Required	Inspection		
					OEM	DTL/TPIA	BERGER/TPI
1	Visual check of material including Make, Model No.	100%	Purchase Specification/Data Sheet	Purchase Order Data Sheet/Technical Specification	P	RW	RW
2	Dimensions, Cable entry, Tag plate marking of detectors and accessories	100%	Purchase Specification/Data Sheet	Purchase Order Data Sheet/Technical Specification GAD/WD	P	RW	RW
3	Functional Check , Alarm check, Power supply	100%	Purchase Specification/Data Sheet	Data Sheet/Technical Specification Test Certificate	P	RW	RW
4	Statutory approval for Flame Detectors, Review of Test Certificate	100%	Purchase Specification/Data Sheet	ATEX/IEC FM PESO/CCOE SIL Test Certificate Witness Report	P P P P P P	R R R R R R	R R R R R R
5	Calibration Certificate of Testing Equipments	100%	Calibration Certificate	Calibration Certificate /Declaration	P	R	R
6	Inspection Certificate / Release note	100%	Purchase Specification	I.R.N.	-	H	P
7	Pre-clearance documentaion	100%	Soft copy and hard copy of Test Certificate and photographic material report	Inspection Report Test Certificate Witness Report	P	N/A	N/A
8	Clearance	By client	Basis of Inspection Report	Test Certificate Witness Report Inspection Report	P	N/A	N/A
9	Dispatch documents	100%	LR/Courier receipt, invoice, packing list, photographic material detail report and Test certificates	Packing List Invoice LR & Inspection Documents	H	N/A	N/A

Legend W: Witness Point H: Hold RW: 10% of Quantity Randomly Per Type Per Lot P: Perform R: Review

ESP Safety Pvt. Ltd.

- 1 Operation Department
- 2 Test and Control Department
- 3 Project Department

ESP Safety Pvt. Ltd. :



Approved By :

 CUSTOMER/END CUSTOMER
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Review BY :

TPI
